

DATA ANALYTICS PROJECT

OCD PATIENT DATASET ANALYSIS

Demographics & Clinical Data | 2013–2022

1,500

Patients

22

Variables

10 Yrs

Study Period

4

Tools Used

Microsoft Excel

MySQL / SQL

Python Jupyter

Power BI



PRESENTATION AGENDA

What we'll cover today

01

Clinical Background

About OCD, obsession & compulsion types, Y-BOCS scale

02

Executive Summary

Dataset overview, key KPIs and high-level findings

03

Demographic Analysis

Gender, age, ethnicity, marital status, education

04

Clinical Symptom Analysis

Obsession & compulsion types, prior diagnoses, symptom duration

05

Y-BOCS Severity Analysis

Score distribution, severity bands, age & gender breakdown

06

Comorbidity & Medication

Depression, anxiety, medication usage by severity

07

Family History & Trends

Genetic factors, diagnosis trends 2013–2022

08

Tools, Findings & Recommendations

Methodology, key insights, and evidence-based recommendations

1

Clinical Background

Understanding OCD — the disorder, scales, and clinical classifications

ABOUT OCD — CLINICAL BACKGROUND

DSM-5 Classification | Global Prevalence | Disorder Profile



What is OCD?

Obsessive-Compulsive Disorder is a chronic mental health condition characterised by persistent, unwanted intrusive thoughts (obsessions) and repetitive behaviours or mental acts (compulsions) performed to reduce distress.

According to the WHO, OCD affects 2–3% of the global population and ranks among the top 10 most disabling illnesses by quality-of-life impact.

2–3%

Global prevalence

Top 10

Most disabling illnesses (WHO)

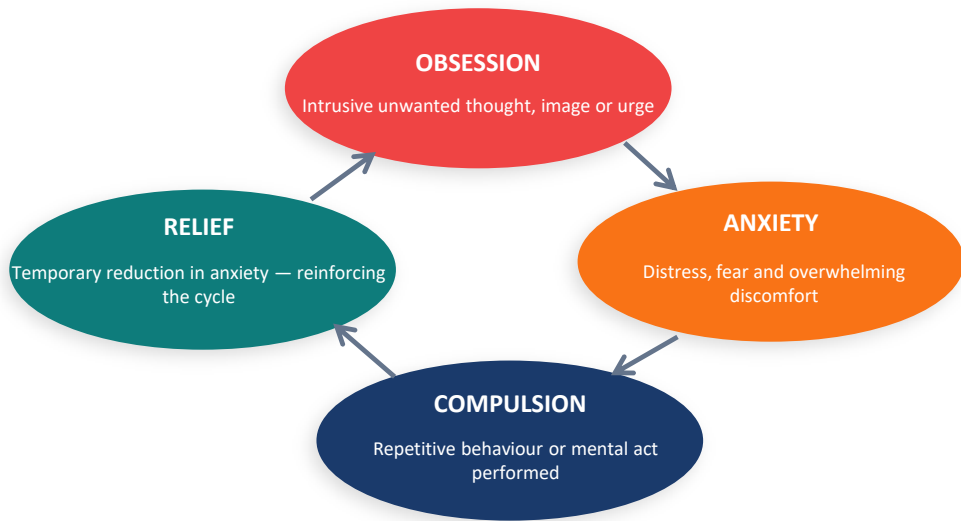
DSM-5

OC & Related Disorders

40–65%

Heritability in adults

The OCD Cycle



Y-BOCS Severity Scale

0–7	8–15	16–23	24–31	32+
Minimal	Mild	Moderate	Severe	Extreme

Dataset Overview

1,500 Patient Records | 22 Variables | 2013–2022 | Pre-cleaned Excel Format

22 Dataset Variables

Patient_ID	Age	Gender	Ethnicity
Marital Status	Education Level	OCD Diagnosis Date	Symptom Duration (mo.)
Obsession Type	Compulsion Type	Y-BOCS Obsessions	Y-BOCS Compulsions
Total Y-BOCS Score	Depression Diagnosis	Anxiety Diagnosis	Medications
Family History of OCD	Previous Diagnoses	Diagnosis Year	Age Group
Severity	Symptom Duration (Yrs)		

Dataset Specifications

Format:	Excel (.xlsx) — 3 worksheets
Rows:	1,500 patient records
Columns:	22 clinical & demographic
Period:	2013 – 2022 (10 years)
Null Values:	None detected
Severity Column:	Computed via IF formula
Duration Column:	Derived from months field
Patients:	All unique (no duplicates)

2

Executive Summary

Key performance indicators and headline findings from 1,500 patient records

EXECUTIVE SUMMARY

Headline findings across 1,500 OCD patients | Excel · SQL · Python · Power BI



1,500

Total Patients



39.67

Avg Y-BOCS Score



10.15 yrs

Avg Symptom Duration



51.5%

With Comorbid Depression



50.7%

Positive Family History

22.2%

Harm-related Obsessions

21.4%

Washing Compulsions

72.5%

Moderate–High Severity

2018

Peak Diagnosis Year

204

Diagnoses in Peak Year

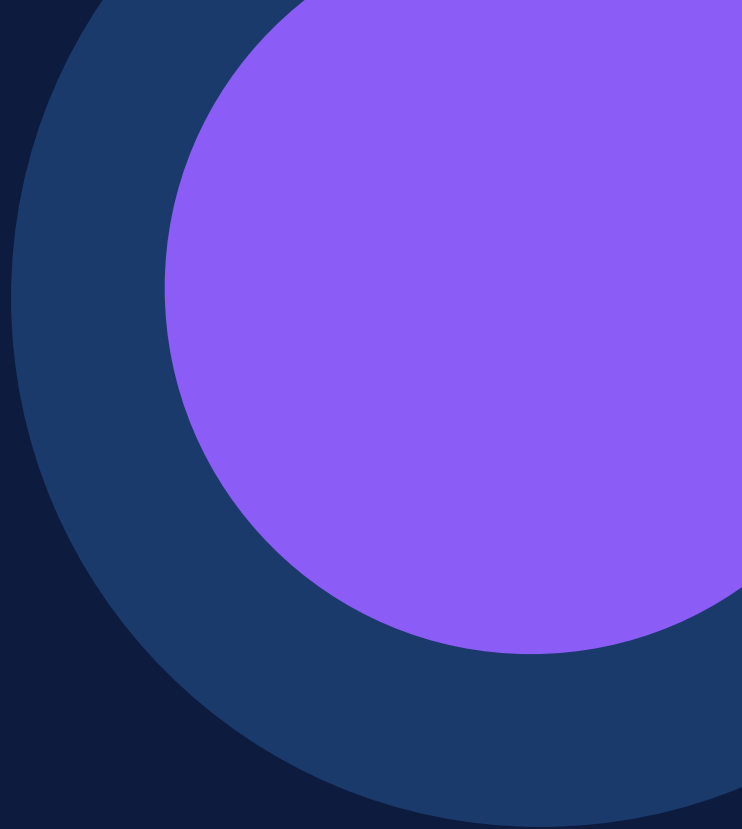
KEY TAKEAWAY

OCD is a chronic, severe, and highly comorbid disorder affecting all demographics equally. The average patient carries 10+ years of symptoms, and over half have comorbid depression — placing enormous burden on healthcare systems and demanding integrated, multi-modal treatment strategies.

3

Demographic Analysis

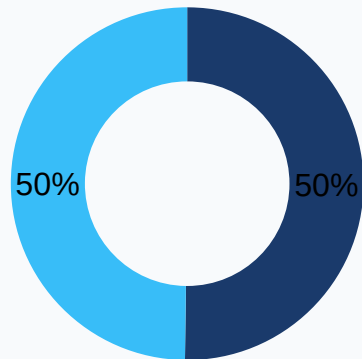
Gender · Age · Ethnicity · Marital Status · Education Level



DEMOGRAPHIC ANALYSIS — GENDER & AGE

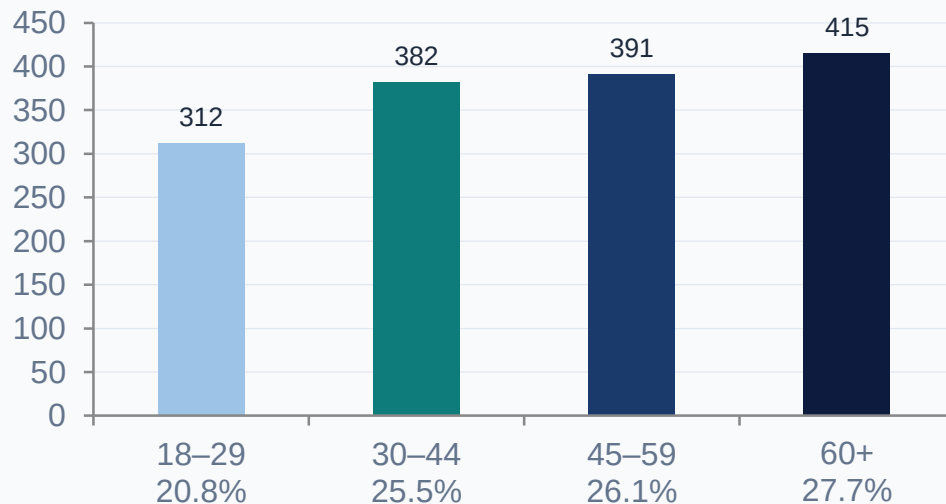
Near-equal gender split | 60+ is the largest cohort

Gender Distribution (n=1,500)



■ Male (50.2%) ■ Female (49.8%)

Age Group Distribution



39.56

Male Avg Y-BOCS

39.79

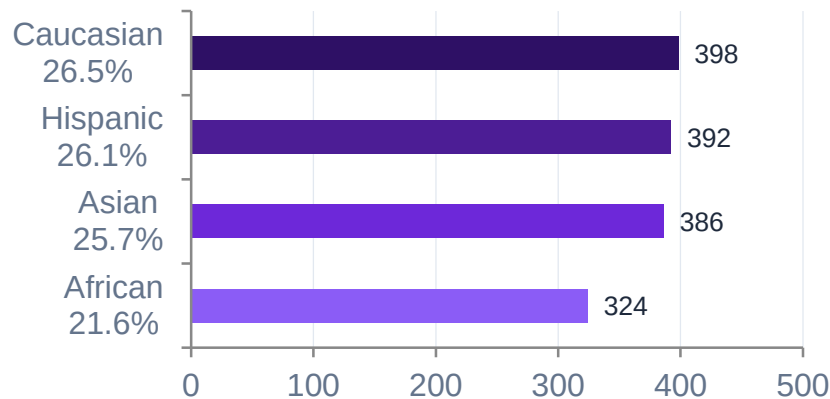
Female Avg Y-BOCS

Insight: The 60+ group is the largest cohort (27.7%), reflecting OCD's chronic accumulation over decades. Gender parity in Y-BOCS scores ($\Delta = 0.23$ pts) confirms severity is gender-neutral.

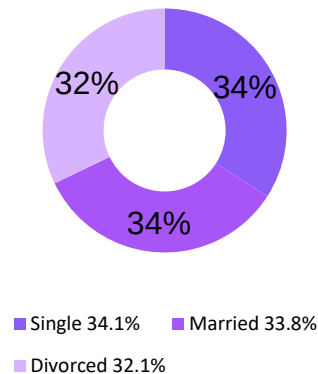
DEMOGRAPHIC ANALYSIS — DIVERSITY PROFILE

Ethnicity · Marital Status · Education Level across 1,500 patients

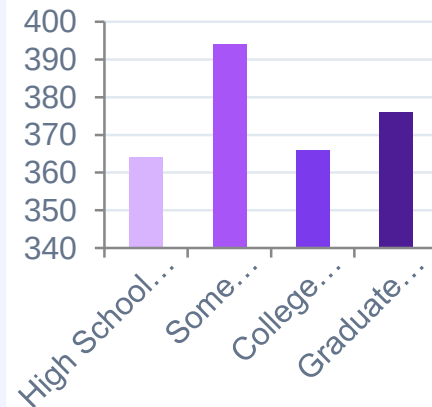
Ethnicity Distribution



Marital Status



Education Level



Broadly representative dataset — 4 ethnicities, Caucasian marginally largest at 26.5%

High divorce rate (32.1%) may warrant investigation of OCD impact on relationships

OCD crosses all education levels with near-equal distribution — no single demographic exempt

4

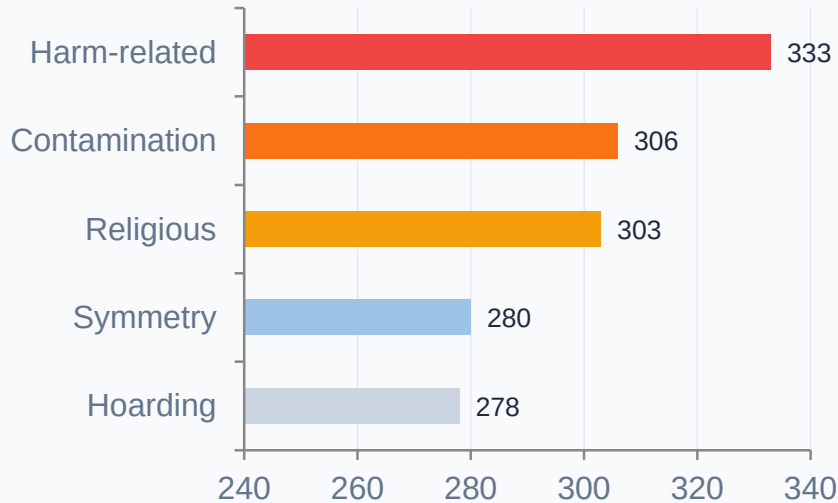
Clinical Symptom Analysis

Obsession types · Compulsion types · Prior diagnoses · Symptom duration

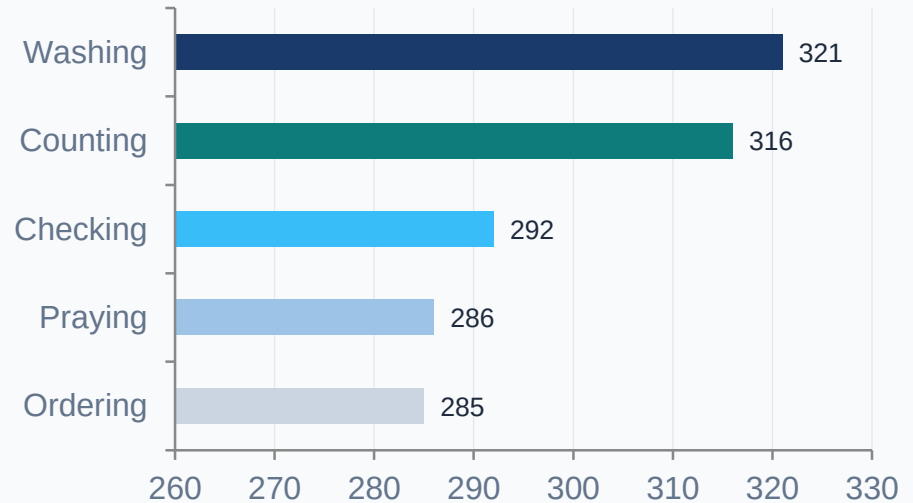
OBSESSION & COMPULSION TYPE ANALYSIS

Distribution of primary symptom subtypes across 1,500 patients

Obsession Types (n=1,500)



Compulsion Types (n=1,500)



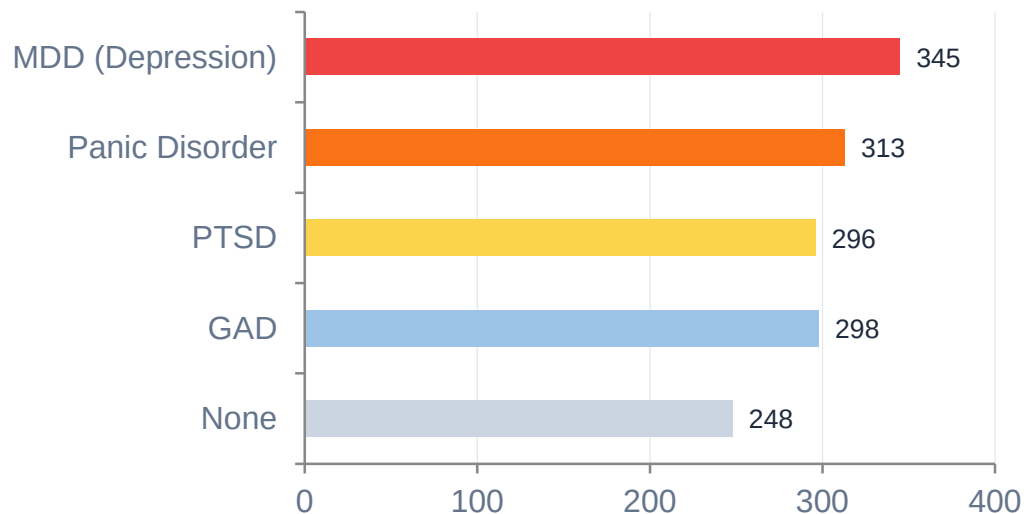
Classic OCD Pairings:

Harm → Checking/Counting | Contamination → Washing | Religious → Praying | Symmetry → Ordering | Hoarding → Collecting rituals

PRIOR DIAGNOSES & SYMPTOM DURATION

Comorbidity history before OCD diagnosis | Chronic symptom burden

Previous Diagnoses Before OCD



83.5% of patients had a prior psychiatric diagnosis — MDD (23%) and Panic Disorder (20.9%) were most common, highlighting OCD's complex comorbidity landscape.

Average Symptom Duration

121.75

months

≈ 10.15 years

OCD Duration Milestones

Year 1

Year 3

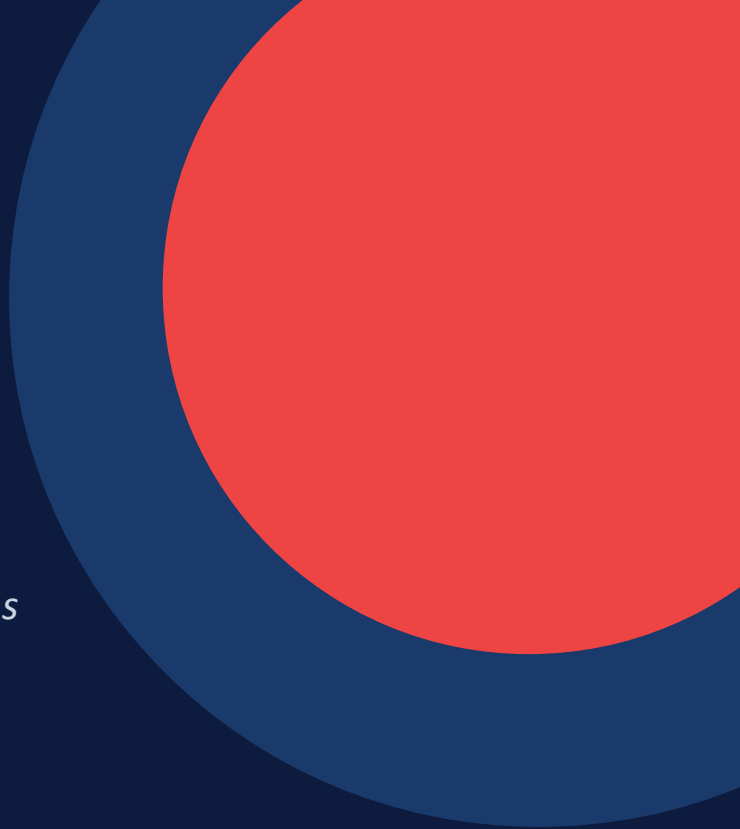
Year 5

Year 10+

5

Y-BOCS Severity Analysis

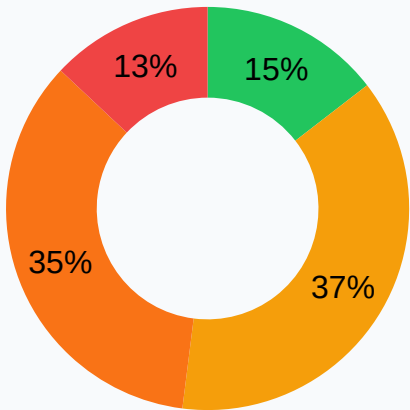
Score distributions · Severity bands · Age & gender breakdowns



Y-BOCS SEVERITY SCORE ANALYSIS

Yale-Brown Obsessive Compulsive Scale | Average 39.67 | Range 1–79

Severity Distribution (n=1,500)



■ Low 14.5% ■ Moderate 37.5% ■ High 35.0% ■ Very High 13.0%

Severity Classification (Dataset-Specific Bands):



1

Minimum

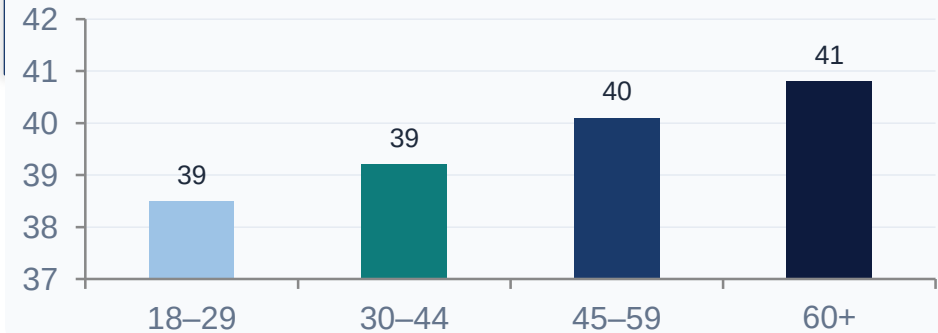
39.67

Average

79

Maximum

Avg Y-BOCS Score by Age Group



6

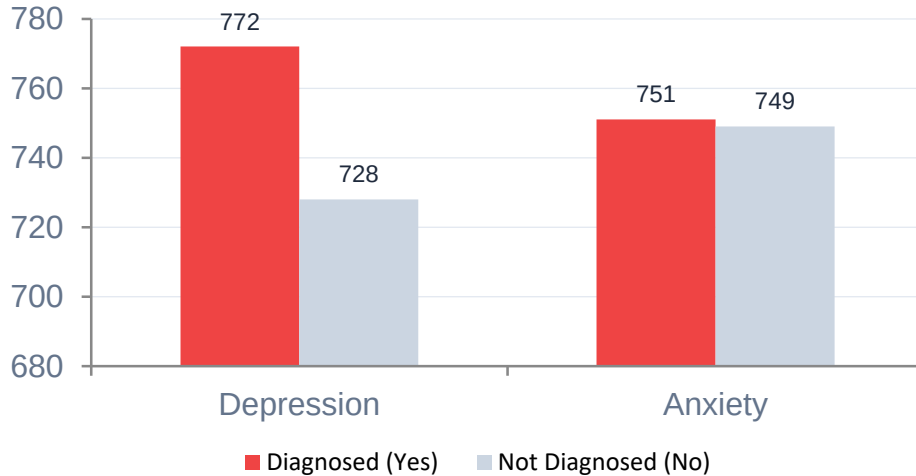
Comorbidity & Medication

Depression · Anxiety comorbidity rates | Medication patterns by severity

COMORBIDITY ANALYSIS — DEPRESSION & ANXIETY

Over half of all OCD patients carry at least one major comorbid condition

Depression & Anxiety Comorbidity



Comorbidity Matrix

D:Yes A:Yes	~390
Highest Complexity	~26%
D:Yes A:No	~382
OCD + Depression	~25.5%
D:No A:Yes	~361
OCD + Anxiety	~24%
D:No A:No	~367

Clinical Alert: 51.5% depression comorbidity far exceeds the expected 25–50% cited in clinical literature. The ~26% with both depression AND anxiety represent the highest-complexity patients requiring multi-disciplinary, dual-diagnosis care pathways.

MEDICATION & TREATMENT ANALYSIS

Pharmacological distribution | Severity-based treatment escalation

Medication Distribution



■ Benzodiazepine 25.7% ■ SNRI 25.1% ■ None 25.7% ■ SSRI 23.5%

376

SNRI

352

SSRI

386

Benzo

386

None

Treatment Pathway by Severity (Clinical Guidelines)

Low

CBT / ERP first-line; pharmacotherapy optional

Moderate

CBT + ERP combined with SSRI (sertraline/fluoxetine)

High

Intensive ERP + SSRI at higher dose; consider clomipramine

Very High

Specialist referral; augmentation strategies; potential hospitalization

Key Observation: Benzodiazepine use equals the unmedicated rate (25.7% each). SSRIs, the first-line recommendation for OCD, account for only 23.5% — suggesting possible over-reliance on anxiolytics rather than OCD-specific pharmacotherapy.

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Family History & Diagnosis Trends

Genetic analysis · Year-on-year diagnosis patterns 2013–2022

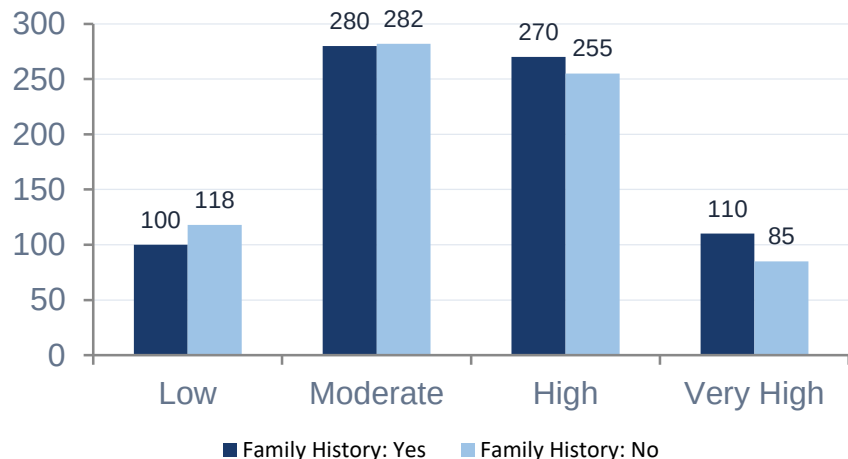


FAMILY HISTORY & DIAGNOSIS YEAR

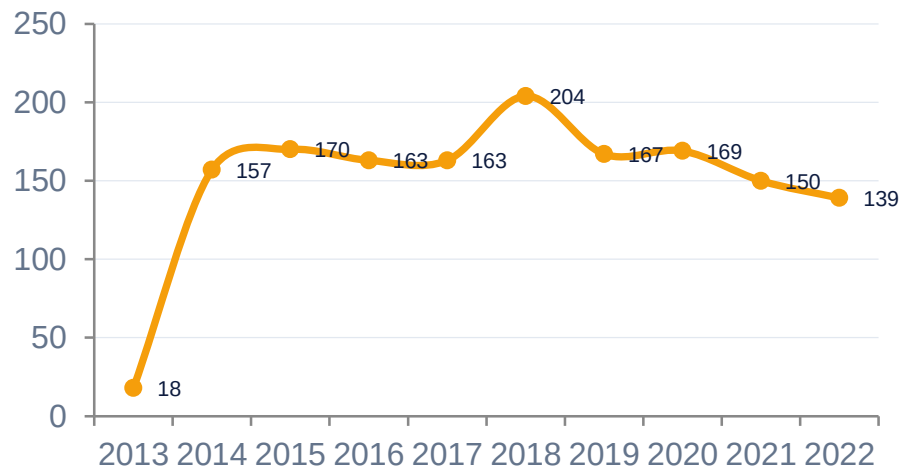
TRENDS

Heritability analysis | OCD diagnosis rates 2013–2022 | 2018 peak

Family History vs. OCD Severity



OCD Diagnosis Trend (2013–2022)



50.7%

Positive Family History

760 of 1,500 patients

40–65%

OCD Heritability

Consistent with genetic research

+3%

Very High Severity

In family history positive group

Trend Analysis

2018 peak (204 Dx) likely reflects awareness campaigns. 2020 remained stable (169) despite COVID-19 — telehealth adoption maintained access. 2021–22 decline may reflect partial data or post-pandemic normalization.

TOOLS & METHODOLOGY PIPELINE

9-Step analytical process across 4 industry-standard platforms



Steps 1-2

Microsoft Excel

Data storage, pivot tables, frequency distributions, severity classification via IF formula, derived columns.



Steps 3-4

MySQL / SQL

CREATE DATABASE, normalised ocd_patients table, GROUP BY aggregations, COUNT/AVG queries, patient_summary VIEW.



Steps 5-6

Python (Jupyter)

Pandas, NumPy, Matplotlib, Seaborn — EDA, univariate/bivariate analysis, correlation, SciPy statistical tests.



Steps 7-8

Microsoft Power BI

Interactive dashboard with dynamic slicers (Gender, Age, Ethnicity, Severity) — bar, line, donut charts + KPI cards.

1. Import

2. Clean

3. Univariate

4. Bivariate

5. Stats

6. Trends

7. Segment

8. Visualise

9. Report

EVIDENCE-BASED RECOMMENDATIONS

Clinical · Healthcare System · Research priorities emerging from this analysis



Clinical Practice

- 1 Universal depression screening at first OCD presentation (51.5% comorbidity rate)
- 2 Develop age-specific protocols, especially for 60+ — highest severity & largest group
- 3 Integrate genetic counselling & family history assessment into OCD intake procedures
- 4 Review benzodiazepine use — not a first-line OCD pharmacotherapy; prefer SSRIs
- 5 Train clinicians across all 5 obsession subtypes equally — none dominates sufficiently



Healthcare System

- 1 Early intervention for 18–29 age group to prevent chronic symptom consolidation
- 2 Replicate 2018 conditions — awareness campaigns & expanded diagnostic access
- 3 Triple-burden pathway for OCD + Depression + Anxiety (~26% of this cohort)
- 4 Leverage telehealth — COVID-19 evidence shows diagnostic stability via remote access
- 5 Standardise reporting across years to address 2021–22 data decline uncertainty



Research Priorities

- 1 Longitudinal follow-up to track Y-BOCS trajectory with treatment over time
- 2 Investigate obsession–compulsion pair interactions and their treatment outcomes
- 3 Survival analysis on symptom duration — identify predictors of prolonged OCD
- 4 Add socioeconomic variables (income, insurance, geography) as key confounders
- 5 Explore severity levels of comorbid conditions, not just binary Yes/No presence

KEY FINDINGS & INSIGHTS

01

Gender-Neutral Severity

50.2% Male / 49.8% Female split; Y-BOCS diff = 0.23 pts — clinically insignificant. Gender-specific protocols unwarranted.

03

Harm-related Obsessions Dominate

22.2% prevalence. Requires specialized risk assessment protocols. Closely linked to Checking & Counting compulsions.

05

72.5% Moderate–High Severity

Major clinical burden. This population requires structured, evidence-based ERP and/or pharmacotherapy interventions.

07

50.7% Positive Family History

Supports genetic component of OCD (40–65% heritability). Positive FH → 3% higher Very High severity rate.

09

2018 Peak — 204 Diagnoses

Highest diagnosis year. Likely linked to mental health awareness campaigns and expanded diagnostic access.

02

60+ is largest cohort (27.7%) with highest avg Y-BOCS (40.8). Reflects chronic, decades-long OCD accumulation.

04

Washing Compulsions #1

21.4% of patients — classic Contamination–Washing pairing. Public health implications around hygiene behaviours.

06

51.5% Comorbid Depression

Highest single-finding impact. Far exceeds literature benchmarks. Demands integrated dual-diagnosis psychiatric pathways.

08

10+ Year Symptom Duration

Avg 121.75 months before dataset endpoint — chronic under-treatment evident. Early screening programmes are essential.

10

Benzodiazepine = Unmedicated Rate

Both at 25.7% — raises pharmacotherapy appropriateness concerns. SSRIs at only 23.5% vs. guidelines recommendation.

STUDY LIMITATIONS

Acknowledging constraints of this analytical dataset



Synthetic / Anonymised Dataset

Findings cannot be directly extrapolated to real-world clinical populations without external validation on confirmed patient records.



Primary Medication Only

Only one medication category is recorded per patient — polypharmacy, dosage information, and augmentation strategies are not captured.



Partial Year Data (2013)

Only 18 patients in 2013 — representing a partial collection year. This distorts long-term trend analysis if treated as a full data point.



Binary Comorbidity Variables

Depression and anxiety are recorded as Yes/No only. Severity levels for these comorbidities are absent, limiting nuanced analysis.



Cross-Sectional Design

No longitudinal outcome data. Treatment responses, remission rates, and relapse patterns cannot be assessed from this dataset.



Missing Socioeconomic Factors

Income, insurance coverage, geography, and access to care are absent — key confounders in mental healthcare access and outcomes research.

CONCLUSION

A complete multi-tool data analytics pipeline applied to 1,500 OCD patients across a decade.



39.67

Avg Y-BOCS

Moderate-to-High severity across the full cohort

51.5%

Depression

Comorbidity — the single most impactful clinical finding

10.15 yrs

Symptom Duration

Chronic under-treatment evident — early intervention needed

72.5%

Mod-High Severity

Active, structured intervention required for most patients